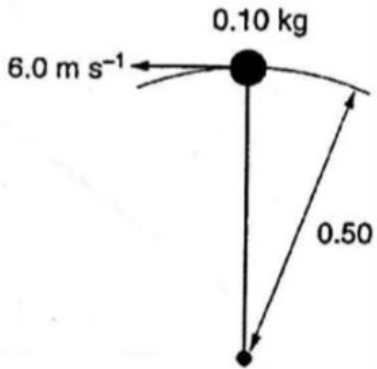
7 A ball of mass 0.10 kg is attached to a string and swung in a vertical circle of radius 0.50 m, as shown. Its speed at the top of the circle is 6.0 m s^{-1}.



What is the tension in the string at this moment?

A 0.98 N **B** 6.2 N **C** 7.2 N **D** 8.2 N

Ans: B

At the **top** of the vertical circle, both the tension T and the weight W a
downward toward the centre of the circle, providing the centripetal fo

$$T + W = \frac{mv^2}{r}$$

Substituting values

centripetal force:

$$\frac{mv^2}{r} = \frac{0.10 \times 6.0^2}{0.50} = \frac{0.10 \times 36}{0.50} = 7.2 \text{ N}$$

weight:

$$W = mg = 0.10 \times 9.81 = 0.98 \text{ N}$$

solving for tension:

$$T = \frac{mv^2}{r} - W = 7.2 - 0.98 = 6.22 \text{ N} \approx 6.2 \text{ N}$$